Department Of Mechanical Engineering

Aliah University

Sub- Engineering Graphics, Paper code-CENUGES-01

Full marks-40 Time- 1.30hrs

Attempt any four **SET A** **10x4**

1. Construct a plain scale to show meter and decimeters and long enough to measure up to 6 meters. R.F of the scale is 1/50. Show on scale, a distance of 4metres and 7 decimeters.

2. The distance between two places is 2.5 km on inspection of the road map, its equivalent distance measures 5cm. Draw a diagonal scale to read 50m minimum. Show on it a distance of 6350m

3. A 55mm long line AB, has its end A at a distance of 20mm in front of the V.P. the line is perpendicular to V.P and 40mm above H.P, draw the projection of the line.

4. A ball thrown from the ground level reaches a maximum height of 6m and travel horizontal distance of 14m from the point of projection. Trace the path of the ball.

5. Draw the involute of a square of side 35mm.

"U.G." END SEMESTER EXAMINATION – JANUARY, 2025
1ST YEAR (1ST SEMESTER) – MECHANICAL ENGINEERING
ENGINEERING MECHANICS
MENUGES01

[Full Marks: 80] [Time: 3:00 hrs.]

Assume suitable data, if not provided

A. **Multiple choice questions (*Compulsory*) –** [$15X1 = 15$]

1. If the resulting moment about the door pivot axis at O is 35 N-m (**figure A1**), the force magnitude is –
   a) 16.65 N b) 22.65 N c) 32.65 N d) None of these

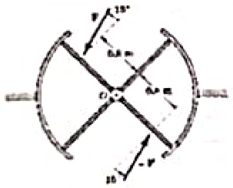

Figure A1

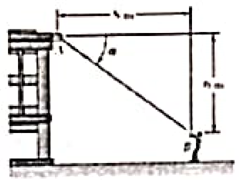

Figure A2

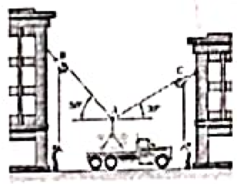

Figure A3

2. A man pulls with a force of 300 N on a rope attached to the top of a building, as shown in figure A2. This force can be written in vectorially as –
   a) $\vec{F} = -240\hat{\imath} + 180\hat{\jmath}\ N$ b) $\vec{F} = 240\hat{\imath} - 180\hat{\jmath}\ N$
   c) $\vec{F} = 240\hat{\imath} + 180\hat{\jmath}\ N$ d) $\vec{F} = -240\hat{\imath} - 180\hat{\jmath}\ N$
3. The values of tension ropes T_{AB} and T_{AC} under equilibrium conditions (figure A3) become –
   a) $T_{AB} = 547\ N$ and $T_{AC} = 480\ N$ b) $T_{AB} = 647\ N$ and $T_{AC} = 380\ N$
   c) $T_{AB} = 647\ N$ and $T_{AC} = 480\ N$ d) $T_{AB} = 547\ N$ and $T_{AC} = 380\ N$

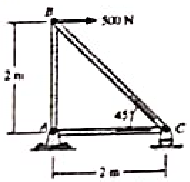

Figure A4

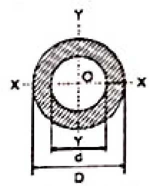

Figure A5

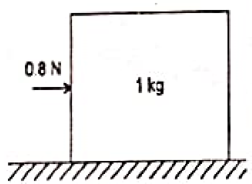

Figure A6

4. In figure A4, the member AC is in –
   a) Tension b) Compression c) Neutral d) None of these
5. The moment of inertia in a hollow section shown in **figure A5** with respect to X-X' for 20 mm external and 10 mm inner diameters becomes –
   a) 8469.72 mm^4 b) 6472.15 mm^4 c) 9363.10 mm^4 d) 7363.10 mm^4
6. A 1-kg block rests on a surface with its static and dynamic coefficient of friction of 0.31 and 0.21, respectively. A 0.8 N force is applied to the block, as shown in **figure A6**. The friction force becomes –
   a) 2N b) 3N c) 4N d) 5N
7. A smooth cylinder lying on a ________ surface is in neutral equilibrium.
   a) Concave b) Convex c) Horizontal d) Zig-Zag
8. A thin disc and a thin ring, both have mass M and radius R. Both rotate about axes through their center of mass and are perpendicular to their surfaces at the same angular velocity. Which of the following is true?
   a) The ring has higher kinetic energy
   b) The disc has higher kinetic energy
   c) The ring and the disc have the same kinetic energy
   d) Kinetic energies of both the bodies are zero, since they are not in linear motion
9. If two forces each equal to P in magnitude act at right angles, their effect may be neutralized by a third force acting along their bisector in opposite direction whose magnitude is –
   a) 2P b) 0.5P c) 0.707P d) 1.414P
10. A framed structure with joints j is said to be perfect, if it contains members equal to –
    a) $2j - 3$ b) $3j - 2$ c) $2j - 1$ d) $j - 2$
11. Forces $5i + 6j + 7k\ N$ and $-8i - 9k\ N$ are acting at a point. The resultant force will be of magnitude –
    a) 5 N b) 7 N c) 9 N d) 11 N
12. A body is said to be in stable equilibrium state, if the second derivative of its total potential energy is ________ zero.
    a) equal to b) greater than c) less than d) none of these
13. The position coordinate of a particle which is confined to move along a straight line is given by $s = t^3 - 12t + 6$, where s is measured in meters from a convenient origin and t is in seconds. The acceleration of the particle at $t = 4s$ is –
    a) 24 m/s^2 b) 48 m/s^2 c) 66 m/s^2 d) 88 m/s^2
14. If the expression of linear momentum is $\vec{G} = \frac{3}{2}(t^2 + 3)\hat{\jmath} - \frac{3}{4}(t^4 - 4)\hat{k}$, then the force in Newton will be –
    a) $t\hat{\jmath} - 2t^2\hat{k}$ b) $2t\hat{\jmath} - 2t^3\hat{k}$ c) $3t\hat{\jmath} - 3t^3\hat{k}$ d) $3t\hat{\jmath} - 2t^3\hat{k}$
15. Four particles A, B, C and D of mass $\frac{m}{2}$, m, 2m and 4m have the same momentum, respectively. The particle with maximum kinetic energy is –
    a) A b) B c) C d) D

B. **Short questions (*Answer any five questions*):** [$5X5 = 25$]

1. The frame shown in figure B1 supports part of the roof of a small building. Knowing that the tension in the cable is 150 kN, determine the reaction at the fixed end E. [5]

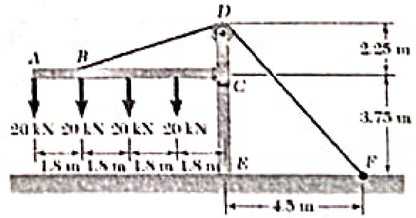

Figure B1

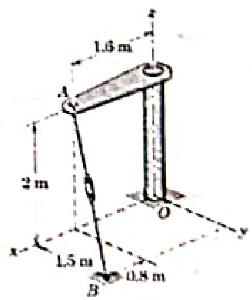

Figure B2

2. The turnbuckle is tightened until the tension is cable AB is 3.4 kN. Determine the moment about point O of the cable force acting on point A (figure B2). [5]

3. Compute the volume V and mass of the solid generated (material is steel) by revolving the right triangle about the z-axis (figure B3) through 180°. (Consider the density of steel is – 7830 kg/m³) [3+2=5]

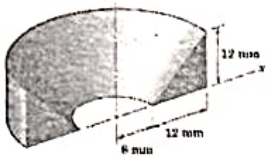

Figure B3

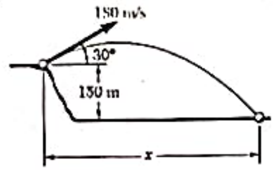

Figure B4

4. A projectile is fired from the edge of a 150 m cliff with an initial velocity of 180 m/s at an angle 30° with the horizontal as shown in figure B4. Neglecting air resistance, find the horizontal distance from the gun to the point where the projectile strikes on the ground. [5]

5. A 75-kg man stands on a spring scale in an elevator. During the first 3 seconds of motion from rest, the tension T in the hoisting cable is 8300 N. Find the reading R of the scale in kg during this interval. (*no figure*) [5]

6. A 120-gm baseball is pitched with a velocity of 24 m/s toward a batter. After the ball is hit by the bat B, it has a velocity of 36 m/s in the direction shown in figure B6. If the bat and ball are in contact for 0.015s, determine the average impulsive force (magnitude and direction) exerted on the ball during the impact. [5]

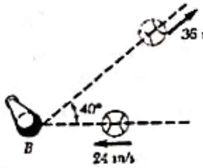

Figure B6

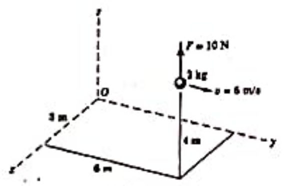

Figure B7

7. A small sphere has the position and velocity indicated in the figure B7 and is acted upon by the force F. Determine the angular momentum H_0 about point O and the time derivative $\dot{H}_0$. [3+2=5]

C. Long questions (*Answer any four questions*): $[4X10 = 40]$

1. Determine the resultant (force and couple) of the four forces and one couple which act on the plate shown in figure C1 [5+5=10]

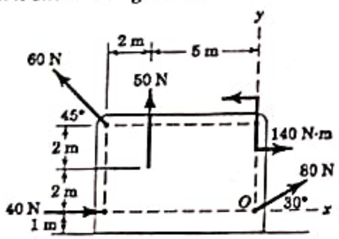

Figure C1

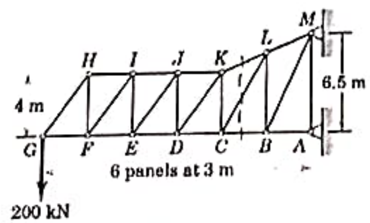

Figure C2

2. Calculate the forces induced in members KL, CL, and CB by the 200-N load on the cantilever truss (figure C2). [3+3+4=10]

3. Each of the two uniform hinged bars has mass m and length l, and is supported and loaded as shown in figure C3. For a given force P, determine the angle θ for equilibrium. Find the angle generated by the system under 50 N applied force. (Consider the mass of each bar is 5 kg of length 0.5m. [8+2=10]

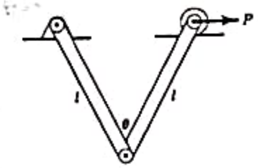

Figure C3

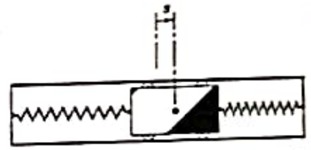

Figure C4

4. The spring-mounted slider moves in the horizontal guide with negligible friction and has a velocity v_0 in the s-direction as it crosses the mid-position, where $s = 0$ and $t = 0$ (figure C4). The two springs together exert a retarding force to the motion of the slider, which gives it an acceleration proportional to the displacement but oppositely directed, i.e., $a = -k^2 s$, where $k =$ constant. Determine the expressions for the displacement s and velocity v as functions of time t. [5+5=10]

5. Aircraft B has a constant speed of 150 m/s as it passes the bottom of a circular loop of 400-m radius as shown in figure C5. Aircraft A flying horizontally in the plane of the loop passes 100 m directly below B at a constant speed of 100 m/s. Determine the instantaneous velocity and acceleration which A appears to have to the pilot of B, who is fixed to his rotating aircraft. [4+6=10]

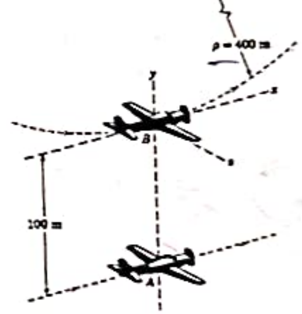

Figure C5

6. A 120-kg block is suspended from an inextensible cable that is wrapped around a drum with a 0.4-m radius that is rigidly attached to a flywheel shown in figure C6. The drum and flywheel have a combined centroidal moment of inertia of $\bar{I} = 16kg.m^2$. At the instant shown, the velocity of the block is 2 m/s directed downward. Knowing that the bearing at A is poorly lubricated so that the bearing friction is equivalent to a couple M of magnitude 90 N.m, determine the velocity of the block after it has moved 1.25 m downward. [10]

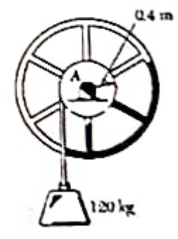

Figure C6

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B.Tech. Examination-2024-25
Electronics and Communication Engineering Department
(Odd Semester Regular and Supplementary)
Course Title - Basic Electronics Engineering   Course Code - ECEUGES01

Full Marks : 80   **(Answer any five, Symbols have their usual meaning)**   **Time-3 hrs**

1. (a) Compare between metal, semiconductor and insulator concerning their band diagram. — 4
   (b) Explain electron-hole pair generation in intrinsic Si semiconductor at room temperature. Can it behave as an insulator at room temperature? Explain — 3+2 : 16
   (c) How n-type semiconductor is formed ? — 3
   (d) Explain why in n-type semiconductor the Fermi level energy E_F is close to the conduction band edge E_C of the band diagram. — 4

2. (a) Define Fermi-level. — 2
   (b) Explain the rectangular shape of Fermi function at 0K. — 6
   (c) Show that Fermi function is symmetrical about E_F i.e. $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$ — 8

3. (a) Show that equilibrium concentrations of electrons (n_0) in a semiconductor can be expressed as :- — 5
$$n_0 = N_c e^{-\left(\frac{E_C - E_F}{KT}\right)}$$
   (b) Show that intrinsic carrier concentration under equilibrium is constant and can be expressed as :- $n_i = \sqrt{N_C N_V}\, e^{-\frac{E_g}{2KT}}$. — 5
   (c) Explain the formation of depletion region and barrier potential V_0 in a p-n junction diode under equilibrium. — 6

4. (a) Explain with diagram the forward bias and reverse bias characteristics of p-n junction diode. — 4+4
   (b) Write down Shockley diode equation for p-n junction diode. — 2
   (c) Explain the avalanche and Zener breakdown mechanism in a p-n junction diode. — 3+3

5. (a) Draw and explain the output waveform of the circuit shown in Fig. 1. — 5
   (b) Draw and explain the output waveform of the circuit shown in Fig. 2. — 5
   (c) What will be the output waveform of the circuit shown in Fig. 3. Explain. — 6

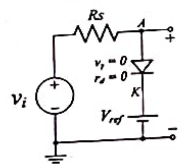

Fig. 1

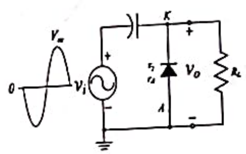

Fig 2

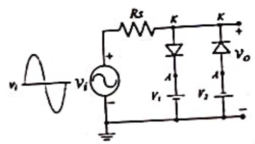

Fig.3

6. (a) Explain the operation of full wave rectifier circuit using bridge network. — 6
   (b) Show that the output DC voltage (V_{dc}) and ripple factor of full wave rectifier output is : (i) $V_{dc} = \frac{2V_m}{\pi}$ (ii) $V_{rms} = \frac{V_m}{\sqrt{2}}$ (ii) $\Gamma = 48.3\%$ — 3+3+4

7. (a) Explain different current components in a p-n-p transistor in active region with a suitable diagram. — 5
   (b) Establish the relation between current amplification factors α and β. — 5
   (c) Briefly explain the CB, CE and CC mode of operation in a transistor. — 6

8. (a) What are the ideal characteristics of OPAMP? — 4
   (b) Derive an expression of voltage gain of a non-inverting amplifier. — 6
   (c) Draw the circuit diagram of a differentiator and find its output voltage expression. — 6

9.(a) Draw and explain the operation of OP-AMP as an inverting amplifier. 4

(b) What is the significance of the virtual ground that exists at the input of the ideal amplifier? 2

(c) What is offset null in IC741? Design and analyse an integrator using an OP-AMP. 2+4

(d) For an OP-AMP having a slew rate, of 2V/μS, what is the ~~maximum~~ voltage gain that can be used when the input signal varies by 0.5V in 10μS? 4

10.(a) Explain the construction and formation of inversion layer in n-channel MOSFET with suitable diagram. 8

(b) Draw and explain the output characteristics of MOSFET. 8

$$\frac{\Delta V_o}{\Delta t} = a \frac{\Delta V_i}{\Delta t}$$

$$\Rightarrow a = \frac{\Delta V_o / \Delta t}{\Delta V_i / \Delta t}$$

$$\frac{\Delta V_o / \Delta t}{\Delta V_i / \Delta t}$$

$$= \frac{2}{0.5/10}$$

$$= \boxed{40}$$

1, 5

2 ↑

3 ↑

4, 1 ↑

ALIAH UNIVERSITY

Odd (Autumn) Examination 2025

Elementary Arabic and Islamic Studies

Course Code: UCCUGAU01

Full Marks: 80 Time: 3:00Hrs.

ARABIC

Q. 1. Attempt any ten questions, each question carries equal marks. **2x10=20**

i. Identify the letters of the following words and split them: يَسْتَكْبِرُ، يُسْتَغْتَبُوْنَ

ii. Read the following words and write the pronunciations in English: يُمْسِكُهُنَّ، فَلَنُحْيِيَنَّهُ

iii. Merge the following letters to make word. ي+س+ت+ط+ي+ع+و+ن ، م+س+ت+ق+ي+م

iv. Identify the following words either definite (المعرفة) or indefinite (النكرة): بَيْتٌ ، اَلْبُرُوْجُ، يَا كَرِيْمُ، جِسْرٌ

v. Identify the following words as they are 'Noun' (اسم) or 'Verb' (فعل) : بِسْمِ اللّٰه، لَنْ نَبْرَحَ، اَلْحَمْدُ، قَدْ نَرَى

vi. How many kinds of sentences are there in Arabic?

vii. Change the Gender of following words: طَبِيْبٌ ، صَفْرَاءُ ، مُهَنْدِسٌ، مُؤْمِنَةٌ،

viii. Write any four demonstrative Pronouns (اسم الإشارة) of Arabic.

ix. How many kinds of Madd are in Arabic? And what are they?

x. What do you Know about *Tanween*? write in short with examples.

xi. Change the Nominal sentence into verbal sentence and Verbal sentence into Nominal Sentence as well. قَرَأَ الطَّالِبُ الكِتَابَ اَلْبِنْتُ رَجَعَتْ مِنَ الْمَدْرَسَةِ

Q. 2. Attempt any two questions, each of them carries equal marks. **10×2=20**

i. Write about Gender in Arabic in details by citing of examples.

ii. Write the Arabic numbers 1-20 in Arabic scripts.

iii. Write down the Moon Letters (الحروف القمرية) and Sun Letters (الحروف الشمسية) in Arabic.

Islamic Studies

Answer any four Questions: **10x4=40**

1. Write a short note about the life of Prophet Muhammad (PBUH).
2. How Many Pillars are there in "Islam"? Write in brief.
3. What is Tawheed? Define its types with example.
4. What do You Know about Zakat? To whom Zakat is Farz? Write a short note on it.
5. How Many Pillars of Iman? Write about all of them in short.

End-Semester Examination - 2024 (Autumn)
MATUGBS01
(Engineering Mathematics – I)
Semester – I (B. Tech in CEN, MEN, ECE, EEN, CSE/ BCA)

Full Marks – 80 **Time – 3 Hours**

Notations and symbols have their usual meaning.

Use separate answer sheets for Group – A and Group – B
Group–A (40 marks)

1. Answer all the following questions: $1 \times 5 = 5$

(a) Define convergent sequence in Real number.

(b) Write the definition of oscillatory series.

(c) Define local maxima and local minima of a function.

(d) Write the statement of L'Hospital theorem.

(e) State the relation between the Gamma and Beta function.

2. Answer all the following questions: $1 \times 5 = 5$

(a) The value of the $\lim_{x\to 1} \frac{1-x^2}{x^2+x-2}$ is

(1) -2
(2) $-\frac{2}{3}$
(3) 1
(4) 2

$\frac{\Gamma 4\ \Gamma 7}{\Gamma(4+7)}$

(b) The value of the $\beta(4,7) =$

(1) $\frac{2}{99}$
(2) $\frac{1}{99}$
(3) $\frac{1}{33}$
(4) $-\frac{4}{99}$

$\frac{3!\ 6!}{(3+6)!}$

(c) Find the value of c guaranteed by the Rolle's theorem of $f(x) = 3x^3 - 4x$ in $[-1,0]$

(1) $\frac{2}{3}$
(2) $\frac{4}{9}$
(3) $-\frac{2}{3}$
(4) $-\frac{4}{9}$

(d) If a series $\sum x_n$ is convergent then which of the following always occurs

(1) $\lim_{n\to\infty} x_n = 1$
(2) $\lim_{n\to\infty} x_n = \infty$
(3) $\lim_{n\to\infty} x_n = 0$
(4) $\lim_{n\to\infty} x_n \neq 0$

$\frac{\cancel{(1\times 2\times 3)}(1\times 2\times 3\times 4\times 5\times 6)}{\cancel{1\times 2\times 3}\,[1+4\times 5\times 6)}$

1

$\frac{7\times 2\times 3\times \cancel{1\times 2\times 3\times 4\times 5}}{\cancel{1\times 2\times 3\times 4\times 5}\times 6\times 7\times}$

$\frac{1}{\frac{7\times\cancel{8}\times\cancel{8}\ 3}{4}}$

(c) Which of the following is not correct

(1) $\lim\limits_{n\to\infty} \frac{1}{n}$ is convergent

(2) $\lim\limits_{n\to\infty} \frac{1}{n}$ is divergent

(3) $\lim\limits_{n\to\infty} \frac{1}{n^2}$ is convergent

(4) $\lim\limits_{n\to\infty} \frac{1}{\sqrt{n}}$ is divergent

3. Answer any **SIX** of the following questions: $5\times 6=30$

(a) Show that the below sequence is convergent $4+1$

$\sqrt{3}, \sqrt{3+\sqrt{3}}, \sqrt{3+\sqrt{3+\sqrt{3}}}, \sqrt{3+\sqrt{3+\sqrt{3+\sqrt{3}}}}, \ldots$. Find out where it converges.

(b) Test the convergence of the following series by Cauchy's root test. 5

$$\sum_{n=1}^{\infty}\left(\frac{n}{n+1}\right)^{n^2}$$

(c) Test the convergence of the series 5

$$\frac{2}{1\cdot 3}+\frac{3}{3\cdot 5}+\frac{4}{5\cdot 7}+\cdots$$

(d) Find the value of x for which the function $f(x)=x^3-6x^2+9x+7$ has maximum and minimum. Find the global maximum of and minimum values in $[0,6]$. 5

(e) (i) Show that $\lim\limits_{x\to 0}\cos\left(\frac{1}{x}\right)$ does not exist. $4+1$

(ii) Define the radius of convergence.

(f) Verify the Rolle's theorem of the following function $f(x)=2x^3+x^2-4x-2$, in the interval $-\sqrt{2}\le x\le\sqrt{2}$. 5

(g) Show that the function $f(x)$ is continuous at the points $6, 0$, and -2, where, $f(x)=|x|-|x+2|+|x-6|$. 5

(h) Evaluate by L' Hospital rule, $\lim\limits_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\log x}\right)$. 5

(i) If m is a positive integers, then 5

$\beta(m,n)=\frac{(m-1)!}{n(n+1)(n+2)\cdots(n+m-1)}$.

Group - B (40 marks)

1. Answer all the following questions: $1 \times 5 = 5$

(a) The line $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-0}{-1}$ is perpendicular to the line

(1) $\frac{x}{1} = \frac{y}{-1} = \frac{z}{1}$ (3) $\frac{x-1}{2} = \frac{y-1}{-3} = \frac{z-7}{1}$

(2) $\frac{x-2}{1} = \frac{y-5}{1} = \frac{z-0}{1}$ (4) $\frac{x-2}{0} = \frac{y-4}{1} = \frac{z-0}{-1}$

(b) The distance between two planes $12x-4y+3z+10=0$ and $12x-4y+3z+36=0$ is

(1) 1 (3) 3

(2) 2 (4) 4

(c) The angle between two planes $x+y+1=0$ and $y+z+1=0$ is

(1) 0 (3) $\frac{\pi}{3}$

(2) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$

(d) If $f(x,y,z) = 3x^2y - y^3z^2$, then grad f at $(1,-2,-1)$ is equal to

(1) $12i+9j+16k$ (3) $i+9j+16k$

(2) $-12i-9j-16k$ (4) $i+9j-16k$

(e) For which type of curve can we use Green's theorem?

(1) open curve (3) both for open and closed

(2) closed curve (4) all of above

2. Answer all the following questions: $1 \times 5 = 5$

(a) Convert the point $(7, -7\sqrt{3})$ in to the polar coordinate.

(b) Find out the direction cosine of the straight line $\frac{2x-1}{6} = \frac{2-y}{4} = \frac{2z-3}{24}$.

(c) What do you mean by the curvature of a curve?

(d) What do you mean by the arc length of a curve?

(e) Find out the value of the double integral $\int_0^1 \int_0^1 (x^2)\, dxdy$

3. Answer any **SIX** of the following questions: $5 \times 6 = 30$

(a) (i) If a line makes an angle $\frac{\pi}{3}$ with x-axis, $\frac{\pi}{3}$ with y-axis and $\theta \in (0,\pi)$ with z-axis, then find out the value of θ. $2\frac{1}{2}$

(ii) Find out the nature of the quadratic surface $2x^2+5y^2+3z^2-4x+20y-6z=5$ and hence evaluate its volume. $2\frac{1}{2}$

(b) Find out the image of the point $(-3,8,4)$ with respect to the plane $6x-3y-2z+1=0$. 5

(c) Write down the equation of the line that is perpendicular to both lines given below; hence, find out the distance between the lines given below 5

$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-0}{0} \text{ and } \frac{x-1}{0} = \frac{y-1}{1} = \frac{z-2}{1}$$

(d) Find out the equation of the plane passing through the point $(1,1,2)$ and perpendicular to the planes $x-y+z=1$ and $2x-3y+z=1$. 5

Find out the curvature at any point t of the curve 5

$$F(t) = (\cos t, -\sin t, 2t).$$

Find out the torsion of the curve 5

$$F(t) = (e^t, e^{-t}, \sqrt{2}t).$$

Evaluate 5

$$\iint (x+y)\,dx\,dy$$

over the region enclosed by the circle of radius 2, $y=x$ and the positive x axis.

Evaluate 5

$$\iiint (x+y+z+1)\,dx\,dy\,dz$$

over the region defined by $x \geq 0,\ y \geq 0,\ z \geq 0,\ x+y+z \leq 1$.

Use Green's theorem to evaluate 5

$$\oint_C (3xy^2 - 6y)\,dx + (y - 3xy)\,dy$$

where C is the boundary of the region $x=0,\ y=0,\ 2x+3y=12$.

Use Divergence theorem to evaluate 5

$$\iint_S \vec{F}\cdot d\vec{S}$$

where $\vec{F} = \cos(\pi x)\hat{i} + 2xy^3\hat{j} + (z^2-4x)\hat{k}$ and S is the surface of the box with $-1 \leq x \leq 2,\ 0 \leq y \leq 1$ and $1 \leq z \leq 4$.